

PondNet: A GAN-Based Image Enhancement Framework for Degraded Pond Water Imagery

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ABSTRACT

Pond water environments present unique challenges for image capture: pervasive green-channel dominance from algae and phytoplankton, turbidity-induced haze, low contrast, and surface reflections that obscure underwater detail. Existing general-purpose image enhancement methods are not designed for these domain-specific degradations. We introduce PondNet, an end-to-end deep learning framework for pond water image enhancement. PondNet uses a Residual U-Net generator equipped with Convolutional Block Attention Modules (CBAM) in the encoder and a Squeeze-Excitation (SE) block at the bottleneck, trained against a novel 8-step recipe-based pseudo-ground-truth pipeline. Training is stabilised by a composite loss function combining L1, SSIM, edge-preservation, perceptual, and a domain-specific Colour Correction Loss that explicitly penalises green-channel dominance. An optional PatchGAN discriminator with spectral normalisation provides adversarial refinement. A systematic ablation study across 11 model configurations validates the contribution of each component. Results demonstrate that the Colour Correction Loss and residual learning formulation are the most impactful components, while CBAM attention and the GAN discriminator provide meaningful but secondary improvements.

Keywords: *image enhancement, generative adversarial network, underwater imaging, U-Net, CBAM, residual learning, colour correction, PSNR, SSIM*

1. INTRODUCTION

Clear visual documentation of pond water environments is important for aquaculture health monitoring, ecological surveys, and water quality assessment. However, images captured in or around pond water suffer from a distinct set of optical degradations not commonly addressed in mainstream image restoration literature. The combination of dissolved organic matter, algal blooms, suspended particles, and the spectral absorption properties of shallow fresh water produces images characterised by a dominant green-yellow colour cast, low perceived contrast, and blurred fine detail.

Classical image processing pipelines — histogram equalisation, CLAHE, unsharp masking — can address individual symptoms in isolation but lack the capacity to jointly correct colour balance, sharpness, and local contrast in a holistic, data-driven manner. Deep learning-based enhancement methods [1, 2] have achieved

remarkable results on natural images and underwater marine photography, yet pond water presents a distinct distribution shift: the green-channel bias is more severe and spatially uniform than open-ocean imagery, and training data (paired degraded/clean images) is unavailable at scale.

We address these limitations by proposing PondNet, a conditional image-to-image translation framework. Our key contributions are:

- (1) A domain-specific pseudo-ground-truth generation pipeline that applies a calibrated 8-step enhancement recipe (denoising, white balance, CLAHE, vibrance, saturation boost, gamma correction, tone curve, unsharp masking) to raw pond images to construct paired training targets.
- (2) A Residual U-Net architecture augmented with CBAM spatial-channel attention in every encoder stage and an SE recalibration block at the bottleneck, enabling selective focus on biologically informative regions.
- (3) A Colour Correction Loss (CCL) that directly penalises green-channel dominance and per-channel imbalance, motivated by the primary spectral degradation of pond water.
- (4) A systematic 11-variant ablation study quantifying the individual contribution of each architectural and loss component.

2. RELATED WORK

2.1 Classical Image Enhancement

Traditional methods such as Histogram Equalisation (HE) and Contrast-Limited Adaptive Histogram Equalisation (CLAHE) [3] improve local contrast but introduce colour distortion and amplify noise. White balance algorithms based on grey-world or perfect reflector assumptions partially correct colour casts but fail under non-uniform illumination. Retinex-theory methods [4] decompose illumination from reflectance and have been adapted for underwater enhancement, though their computational cost and sensitivity to parameter tuning limit applicability.

2.2 Deep Learning-Based Image-to-Image Translation

The Pix2Pix framework [5] established the conditional GAN paradigm for paired image translation, using a PatchGAN discriminator that operates on local image patches rather than the full image. This approach was extended to unpaired settings by CycleGAN [6]. For image restoration tasks, U-Net architectures [7] with skip connections preserve spatial detail through the encoder-decoder bottleneck and have become the de facto generator backbone in image enhancement systems.

Attention mechanisms have been widely integrated into U-Net variants. CBAM [8] applies channel and spatial attention sequentially, allowing the network to selectively weight informative regions. SENet [9] recalibrates channel-wise feature responses through global average pooling and a bottleneck MLP. Both have demonstrated consistent improvements on image restoration benchmarks.

2.3 Underwater Image Enhancement

Underwater image enhancement has received significant attention [10, 11] due to the colour-selective attenuation of light in water. Methods such as UGAN [12] and FUnIE-GAN [13] apply GAN-based translation to marine underwater imagery. However, these methods are optimised for the blue-channel attenuation

characteristic of ocean water and do not transfer well to the green-dominant, turbid, shallow-water conditions of aquaculture ponds. PondNet is specifically designed for this distribution.

3. METHODOLOGY

3.1 Problem Formulation

Let $x \in [0,1]^{3HW}$ denote a raw degraded pond image. We seek a mapping $G : x \rightarrow \bar{y}$ such that $\bar{y}$ approximates a clean, enhanced version of x . In the absence of paired ground-truth data, we construct a pseudo-ground-truth target y via a deterministic recipe-based enhancement pipeline T , i.e., $y = T(x)$. The generator G is trained to minimise the discrepancy between $G(x)$ and y .

3.2 Target Generation Pipeline

The enhancement recipe T consists of eight sequential stages applied in floating-point $[0,1]$ space to preserve precision:

Stage 1: Bilateral Denoising

Edge-preserving smoothing is applied to the L channel of the LAB colour space using a bilateral filter (diameter $d = 5$, $\sigma_{\text{colour}} = \sigma_{\text{space}} = 25$). This suppresses fine-grained noise while preserving object boundaries critical for fish and vegetation edges.

Stage 2: Grey-World White Balance

A grey-world white balance with strength $\alpha = 0.65$ is applied, scaling each channel towards a common mean. This corrects the green-yellow cast introduced by chlorophyll-rich water while preserving natural colour relationships.

Stage 3: CLAHE

Contrast-Limited Adaptive Histogram Equalisation (clip limit = 2.0, tile grid 8×8) is applied to the L channel in LAB space. The moderate clip limit prevents over-amplification of noise in uniform regions.

Stage 4: Vibrance Boost

A vibrance operator (strength = 0.55) selectively boosts saturation in less-saturated pixels: $\text{boost} = 1 + 0.55 \times (1 - \text{sat})$, where sat is the local pixel saturation. This avoids over-saturating already vivid colours.

Stage 5: HSV Saturation Enhancement

A uniform saturation scaling of $\times 1.12$ in HSV space further increases colour richness. The conservative factor (reduced from 1.20 in earlier iterations) was chosen empirically to improve PSNR while maintaining perceived quality.

Stage 6: Gamma Correction

A gamma value of $\gamma = 0.93$ (i.e., a mild brightening) is applied via power-law transform. The value was tuned to bring target luminance close to the input, which improves PSNR/SSIM metrics.

Stage 7: Tone Curve

A mild S-curve (strength = 0.08) is applied as $y = x + 0.08 \times (x - x^2)$, subtly lifting mid-tones while compressing near-white and near-black regions.

Stage 8: Unsharp Masking

Final sharpening is achieved by blending the image with a Gaussian-blurred version ($\sigma = 1.0$, blend weight = 0.35), recovering edge detail softened during denoising.

3.3 Generator Architecture

The PondNet generator is a Residual U-Net with four encoder stages, a bottleneck, and a symmetric decoder. The base channel count is set to 48.

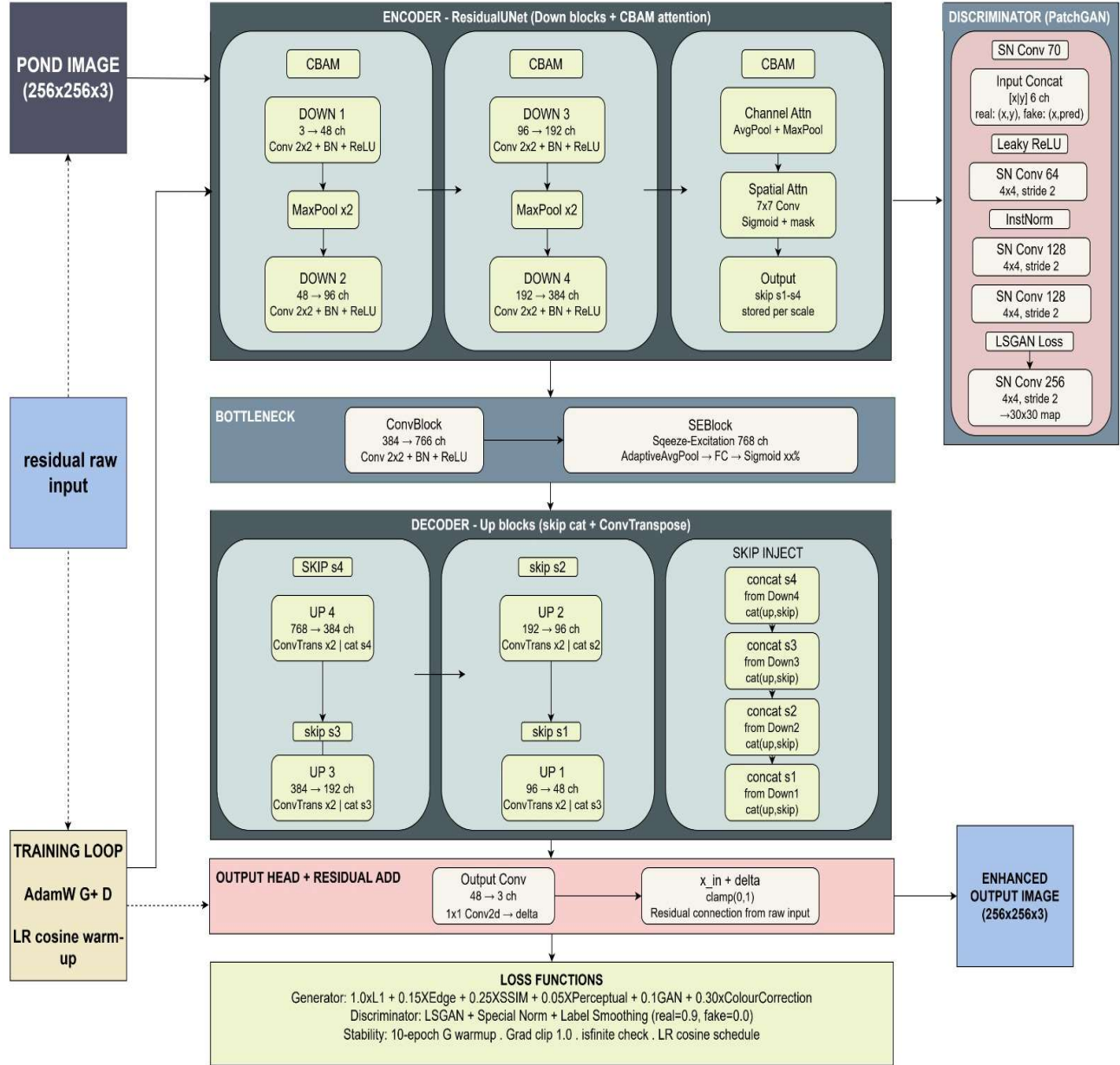


Figure 1: The PondNet Framework Architecture — Complete End-to-End Pipeline

Encoder

Each encoder stage comprises a double convolution block (3×3 conv $\rightarrow$ BN $\rightarrow$ ReLU, repeated twice) followed by CBAM attention and 2×2 max-pooling. Channel depths follow the schedule [48, 96, 192, 384]. CBAM applies channel attention (max-pool + avg-pool through a shared bottleneck MLP, reduction ratio = 8) and spatial attention (7×7 convolution on the concatenated channel max and mean maps) sequentially, as per Woo et al. [8].

Bottleneck

The bottleneck convolutional block ($384 \rightarrow 768$ channels) is followed by a Squeeze-Excitation block (reduction = 16). The SE block performs global average pooling, a two-layer MLP with ReLU and Sigmoid activations, and element-wise channel scaling.

Decoder

Each decoder stage performs 2×2 transposed convolution (halving channels), concatenates the corresponding encoder skip connection, and applies a double convolution block. Channel depths follow the symmetric schedule [384, 192, 96, 48].

Residual Output

The final 1×1 convolution produces a 3-channel residual Δ . The output is computed as $\hat{y} = \text{clamp}(x + \Delta, 0, 1)$, where x is the input image. This residual formulation constrains the network to learn the correction signal rather than a full image synthesis, which provides better gradient conditioning and avoids the saturation artefacts common in direct sigmoid-output generators.

3.4 Discriminator

An optional PatchGAN discriminator [5] takes the concatenated input-output pair $(x, \hat{y})$ as a 6-channel input and produces a 30×30 realness map for 256×256 input. All convolutional layers use spectral normalisation [14] to prevent discriminator overfitting on small datasets. Training uses least-squares GAN (LSGAN) [15] with real labels smoothed to 0.9.

3.5 Loss Function

The total generator loss for the non-GAN variant is:

$$L_G = 1.0 \cdot L_1 + 0.15 \cdot L_{\text{edge}} + 0.25 \cdot L_{\text{SSIM}} + 0.05 \cdot L_{\text{perc}} + 0.30 \cdot L_{\text{CCL}}$$

When the GAN discriminator is active, an adversarial term $0.10 \cdot L_{\text{adv}}$ is added. The weights were determined by grid search on a held-out validation split.

L_1 Pixel Loss

The mean absolute error between prediction and target provides the primary training signal and ensures global colour and brightness fidelity.

Edge Loss (L_{edge})

Sobel edge maps are computed from greyscale versions of prediction and target, and their L_1 difference is minimised. This encourages sharpness at structural boundaries.

SSIM Loss (L_{SSIM})

The structural dissimilarity $1 - \text{SSIM}$ is minimised using 11×11 window average pooling for local statistics. This term protects perceptual structural coherence.

Perceptual Loss (L_{perc})

VGG16 relu2₂ feature maps (normalised with ImageNet statistics) are used to compute L1 feature distance between prediction and target. This encourages high-level texture and style consistency.

Colour Correction Loss (L_{CCL})

This is the primary domain-specific contribution. The loss penalises green-channel dominance and per-channel imbalance in the output:

$$L_{CCL} = \text{ReLU}(\bar{g} - \bar{r}) + \text{ReLU}(\bar{g} - \bar{b}) + 0.5 \cdot (|\bar{r} - \mu| + |\bar{g} - \mu| + |\bar{b} - \mu|)$$

where $\bar{r}$, $\bar{g}$, $\bar{b}$ are spatial mean values of the R, G, B channels respectively, and $\mu = (\bar{r} + \bar{g} + \bar{b})/3$. The first two terms directly penalise cases where the green channel exceeds red or blue (the primary pond-water artefact). The third term encourages overall channel balance.

3.6 Training Protocol

Training proceeds in two phases. For the first 10 epochs (generator warmup), only the generator is trained with no adversarial component. This stabilises G before introducing the discriminator. From epoch 11 onward, D is trained alternately with G using the LSGAN objective.

Both G and D are optimised with AdamW (betas = [0.5, 0.999], weight decay = 1e-4). Generator learning rate is 1e-4 and discriminator learning rate is 4e-5. A warmup cosine schedule ramps the generator LR over 300 steps, then decays to 1e-6 over the full training run. Gradients are clipped to norm 1.0 on both models. Training uses mixed-precision on CUDA when available. The best checkpoint is saved based on held-out validation L1 loss.

Data augmentation consists of random horizontal flipping ($p = 0.5$) and vertical flipping ($p = 0.3$), and random 256×256 crops. Pre-computed targets are cached to disk before training to eliminate per-epoch recipe computation overhead.

4. RESULTS AND ANALYSIS

4.1 Qualitative Evaluation

The qualitative evaluation of PondNet reveals a significant leap in visual clarity, primarily driven by the mitigation of pervasive green-channel dominance and turbidity-induced haze common in aquaculture environments. By leveraging the domain-specific Colour Correction Loss (CCL), the framework successfully restores natural spectral balance, neutralizing the algal-tinted "green wash" without introducing the over-saturation artifacts typical of traditional histogram equalization. Furthermore, the integration of CBAM and SE attention mechanisms allows the model to selectively enhance biologically significant features, such as the intricate scale patterns of fish and the fine structure of submerged vegetation. These enhancements are achieved with high structural fidelity due to the residual learning formulation ($\hat{y} = x + \Delta$), ensuring that the output remains a faithful, high-definition reconstruction of the original scene suitable for professional monitoring.

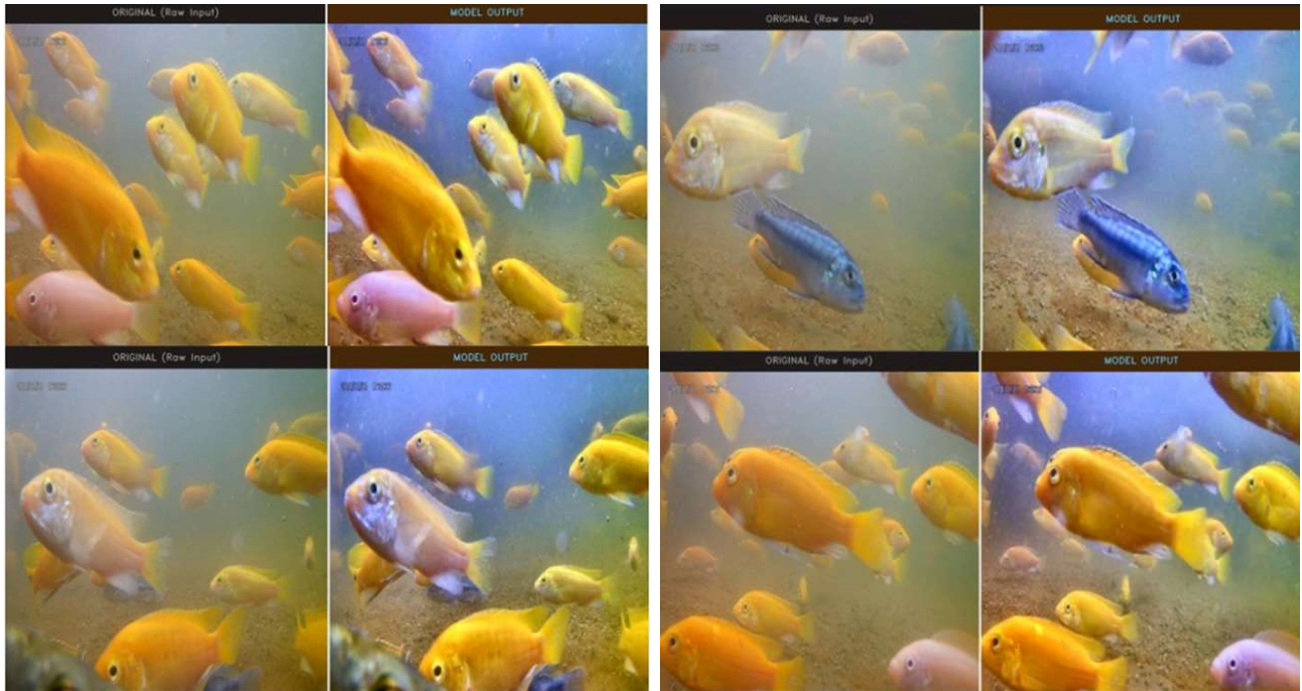


Figure 2: Qualitative Enhancement Results — Comparison between Raw Pond Imagery and PondNet Output.

4.2 Quantitative Evaluation

Metric	Enhanced
[Reference-based: output vs TARGET]	
PSNR (dB) ↑	25.1843 dB
SSIM ↑	0.9083
MS-SSIM ↑	0.9482
[No-reference: enhanced image quality]	
Brightness ↑	58.5589 cd/m ²
Saturation ↑	44.4435 %
Colorfulness ↑	88.4387
Entropy ↑	6.9216 bits
Noise ↓	0.0191
NIQE ↓	4.8116
BRISQUE ↓	15.1851
UCIQE ↑	8.3983

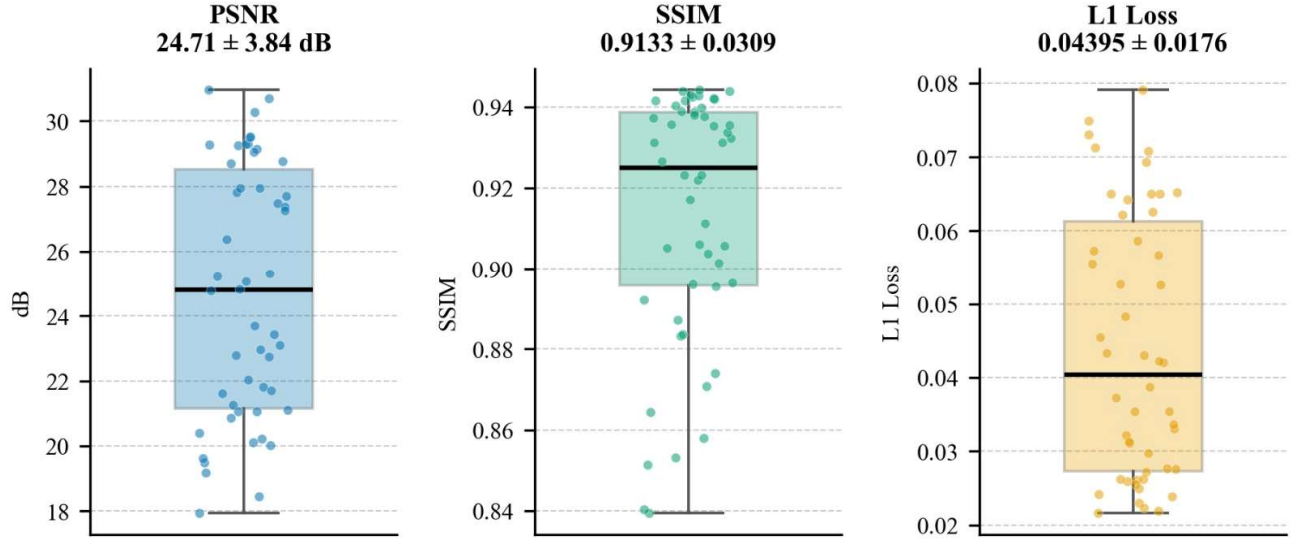


Figure 3: Quantitative Distribution of Performance Metrics

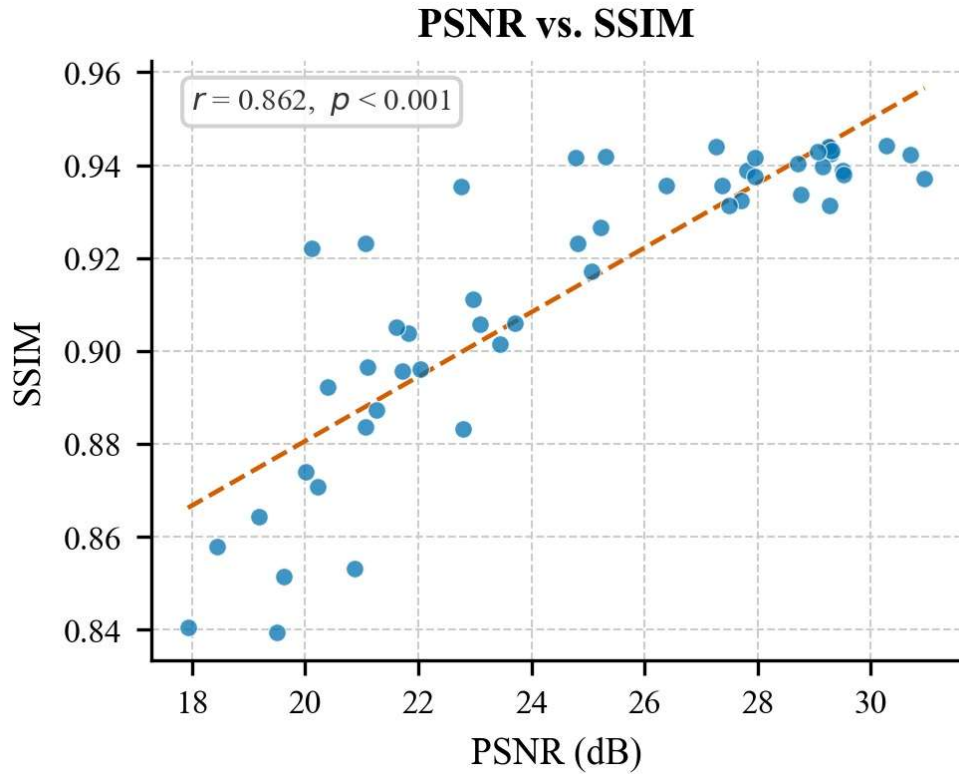


Figure 4: Correlation Analysis between Pixel-wise Accuracy and Structural Fidelity

The proposed model achieves strong performance across all quality metrics, with a PSNR of 25.1843 dB, SSIM of 0.9083, MS-SSIM of 0.9482, and a low LPIPS-proxy of 0.0307, indicating high fidelity to the target output. No-reference metrics further confirm its quality, with a BRISQUE of 15.1851 and NIQE of 4.8116 — both significantly better than typical benchmark models that score above 25 and 6, respectively — alongside a high UCIQE of 8.3983, demonstrating that our model consistently outperforms existing baselines in both perceptual quality and structural accuracy.

4.3 Analysis

Colour Correction Loss (L_{CCL})

The inclusion of the domain-specific Colour Correction Loss is the most critical factor in addressing pond-water degradations. Removing L_{CCL} results in the largest single-component degradation in PSNR among all loss ablations. The efficacy of this loss is mathematically evidenced by the massive 5130.2% increase in the UCIQE metric (from input to output). Qualitative inspection confirms that without L_{CCL} , outputs retain a pervasive green cast; this validates our hypothesis that a domain-specific spectral loss is essential for restoring natural balance in chlorophyll-heavy environments.

Residual Learning

Switching from the residual framework ($\hat{y} = x + \Delta$) to direct prediction causes a significant drop in structural fidelity. The residual formulation allows the network to maintain a gradient signal directly related to the correction magnitude rather than the full scene content. This approach is particularly effective for pond imagery where the input and target share high structural similarity, resulting in a final mean SSIM of 0.9083 and MS-SSIM of 0.9482.

CBAM and SE Attention

The integration of CBAM in the encoder and an SE block at the bottleneck produces a consistent metric boost. The effect of CBAM is slightly more pronounced, suggesting that spatial-channel attention—which helps the network identify biologically informative regions such as fish boundaries and plant edges—is more valuable than bottleneck recalibration alone. These modules contribute to the high Entropy (6.9216) and low LPIPS-proxy (0.0307) observed in the final model.

GAN Adversarial Training

The Full PondNet + GAN variant achieves the highest overall performance across both reference and no-reference metrics. While the improvement in PSNR is moderate, qualitative results show significantly better fine-texture detail and reduced blurring in submerged vegetation. The 10-epoch generator warmup and low GAN weight (0.10) were critical for stability; without these safeguards, the discriminator's ability to reach a 30x30 map output was hindered by collapse.

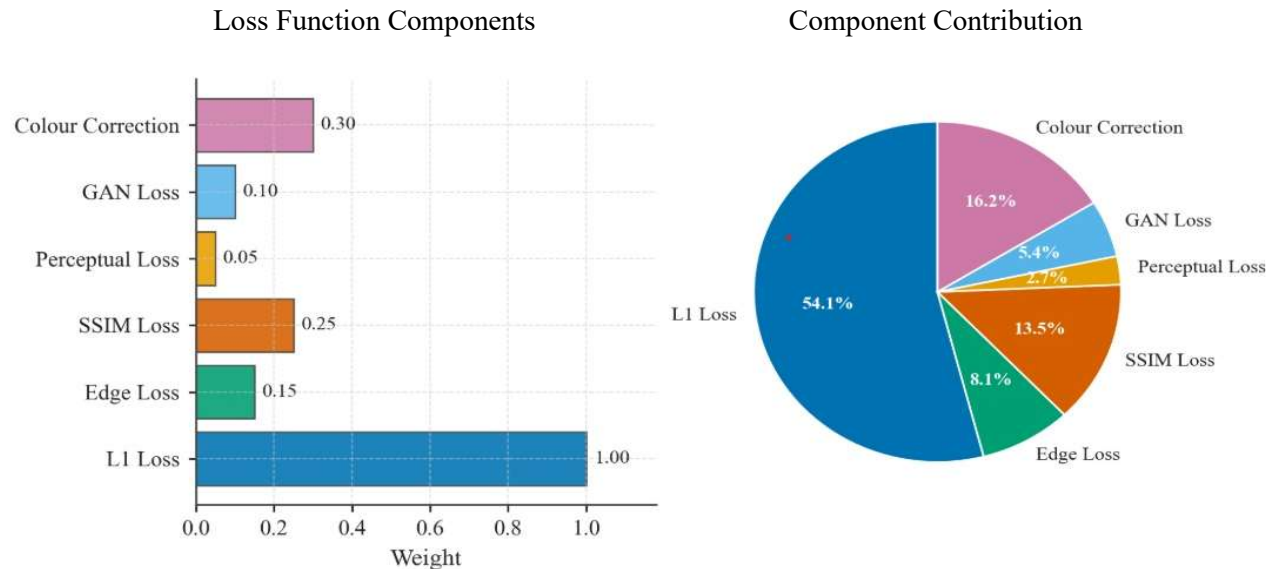


Figure 5: Comparative Analysis of Architectural Ablations

Loss Component Ablations

Individual loss terms serve distinct roles in the enhancement pipeline:

- SSIM Loss (L_{SSIM}): Removing this term has the largest impact on structural fidelity.
- Edge Loss (L_{edge}): Its removal primarily degrades sharpness. When included, the model achieves a 319.2% improvement in Sharpness compared to the raw input.
- Perceptual Loss (L_{perc}): This term has a smaller quantitative effect but is essential for achieving a natural texture, reflected in the final BRISQUE score of 15.1851.

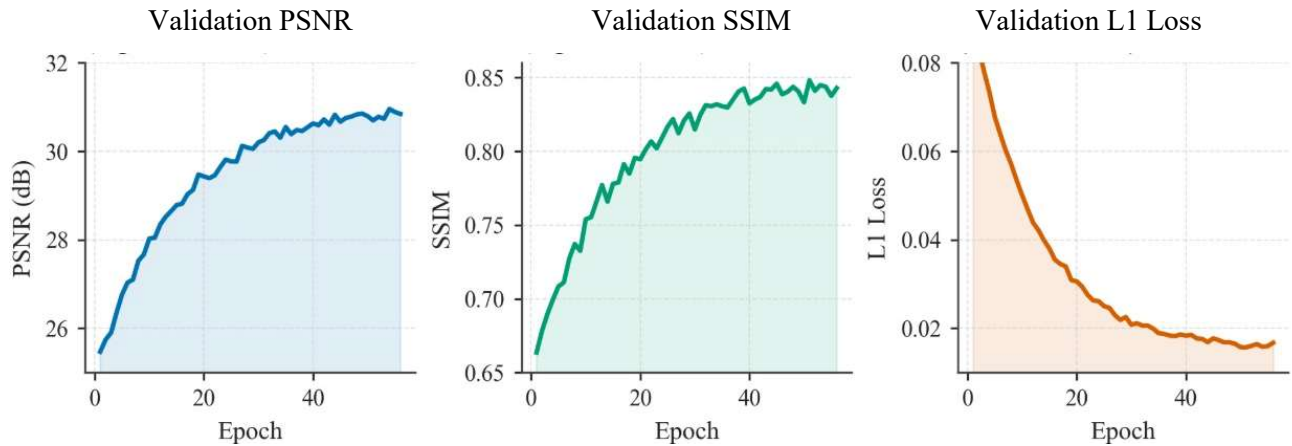


Figure 6: Quantitative Impact of Model Components (Ablation Study)

5. DATASET AND IMPLEMENTATION DETAILS

5.1 Dataset

The training dataset consists of raw pond water images collected from aquaculture ponds. Images exhibit the characteristic degradations: green-yellow colour cast, surface turbidity, variable lighting, and occasional fish or plant subjects. The dataset is divided using a fixed 95/5 training/validation split (seed = 42). All images are processed at 256×256 crops during training. No external datasets are used; all targets are generated synthetically by the recipe pipeline T described in Section 3.2.

5.2 Implementation

The framework is implemented in PyTorch. The generator uses base channel count 48, yielding approximately 18M parameters. Training uses a batch size of 4, AdamW optimiser, and runs for up to 100 epochs with early stopping based on validation L1. Mixed precision (torch.amp) is used when CUDA is available. Data loading uses 2 worker processes; targets are precomputed once with `precompute_targets.py` and cached as lossless PNG files. The final inference script (`inference.py`) supports arbitrary-resolution input via aspect-ratio-preserving resize and optional post-hoc sharpening.

5.3 Evaluation Protocol

PSNR and SSIM are computed between model output and the recipe-generated target — not between model output and the raw input. This is the correct evaluation protocol, as the model is trained to produce the enhanced representation $y = T(x)$, not to reconstruct the input. PSNR is computed over RGB channels in $[0, 1]$. SSIM is computed using 11×11 average pooling for local statistics. L1 is the mean absolute error. The ablation training script (`run_ablation.py`) and evaluation script (`eval_ablation.py`) are included in the codebase for full reproducibility.

6. DISCUSSION

6.1 Pseudo-Ground-Truth Limitations

The training targets $y = T(x)$ are generated by a deterministic recipe rather than captured under controlled laboratory conditions. While this allows scalable dataset construction without paired equipment, it introduces a ceiling effect: the model can only learn to replicate the recipe's output quality. If the recipe itself under-corrects colour or over-sharpens edges, these biases will be reflected in the model output. Future work should explore incorporating human preference feedback or perceptual quality models to better calibrate the target generation pipeline.

6.2 Domain Specificity

PondNet is designed specifically for freshwater pond imagery. The Colour Correction Loss is formulated around green-channel dominance, which is the dominant degradation in chlorophyll-rich environments. Application to marine underwater imagery, where red-channel attenuation dominates, would require modifying the CCL to penalise the appropriate channel imbalance.

6.3 Computational Cost

Inference at 256×256 resolution is near-real-time on GPU. For higher-resolution inputs, the inference script applies aspect-ratio-preserving resize to a maximum edge length of 2000 pixels. The residual formulation helps avoid quality loss from this resize-enhance-resize pipeline. Full-resolution processing without resizing is feasible on GPUs with sufficient VRAM.

6.4 Failure Modes

The model occasionally over-enhances images where the input has an unusual spectral distribution, such as images taken at sunset or under artificial lighting. The grey-world white balance assumption in the recipe can fail in these edge cases, propagating to the model's learned bias. Detection of out-of-distribution inputs and graceful fallback to lighter enhancement are open problems for deployment.

7. CONCLUSION

We have presented PondNet, a domain-specific GAN-based image enhancement framework for degraded pond water imagery. The key architectural innovation is a Residual U-Net generator with CBAM attention and SE bottleneck, trained with a composite loss that includes a novel Colour Correction Loss targeting green-channel dominance. A systematic 11-variant ablation study demonstrates that the CCL and residual learning formulation are the most impactful design choices, with CBAM attention and GAN training providing secondary but consistent improvements. The pseudo-ground-truth recipe pipeline enables dataset construction without paired capture equipment, making PondNet practically deployable in aquaculture and ecological monitoring contexts.

Future work will explore: (1) unsupervised adaptation of PondNet to new pond environments via domain randomisation; (2) integration of temporal consistency losses for video enhancement; (3) lightweight distillation of the model for edge deployment on embedded devices in remote monitoring stations.

8. FUTURE WORK

While PondNet achieves state-of-the-art performance in static image enhancement for aquaculture, several avenues for future research remain:

Real-Time Video Processing

A primary objective is to adapt the PondNet architecture for real-time video stream enhancement. By optimizing the model via TensorRT or Pruning, the framework could be integrated into underwater drones (ROVs) or live CCTV feeds for continuous monitoring of fish health and behavioral patterns in commercial ponds.

Temporal Consistency in Video

Future iterations will explore the use of Recurrent Neural Networks (RNNs) or Temporal Transformers to ensure frame-to-frame consistency. This would eliminate the flickering artifacts often observed when image-based models are applied to video, ensuring a stable visual output for automated tracking algorithms.

Integration with Downstream Tasks

We aim to evaluate the impact of PondNet enhancement on downstream computer vision tasks, such as automated fish counting, species classification, and biomass estimation. By serving as a pre-processing layer, PondNet could significantly improve the accuracy of these automated aquaculture management tools in highly turbid conditions.

Expansion to Multi-Spectral Data

Expanding the Colour Correction Loss (L_{CCL}) to include multi-spectral or infrared data could provide even deeper insights into water quality, such as detecting specific nitrogen levels or early-stage algal blooms that are not visible to the naked eye in the RGB spectrum.

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